

ONTAP MCP Runbook

Claude Desktop (Windows) ↔ ONTAP MCP on Rocky Linux ↔ ONTAP Simulator

1. Purpose

This runbook documents the **end-to-end setup** for integrating **Claude Desktop on Windows** with **NetApp ONTAP MCP** running on a **Rocky Linux VM** in Docker, connected to an **ONTAP Simulator / ONTAP cluster**.

It includes:

- architecture
 - prerequisites
 - Windows setup
 - Rocky Linux setup
 - ONTAP MCP deployment
 - Claude Desktop MCP integration
 - validation steps
 - troubleshooting based on the issues encountered during setup
-

2. Final architecture

``text id="Offuxj" Claude Desktop (Windows) -> local MCP bridge (mcp-remote via Node.js / npx) -> ONTAP MCP server on Rocky Linux VM (Docker, port 8083) -> ONTAP Simulator / ONTAP cluster management LIF -> ONTAP REST API

```
---
```

```
# 3. High-level component roles
```

```
## 3.1 Claude Desktop (Windows)
```

```
Claude Desktop is the **MCP client**.  
It does not talk directly to ONTAP.
```

```
## 3.2 `mcp-remote` (Windows)
```

```
Claude Desktop launches `mcp-remote` locally using Node.js.  
`mcp-remote` acts as a **bridge**:
```

```
- local Claude MCP stdio connection → remote MCP HTTP connection
```

```
## 3.3 ONTAP MCP server (Rocky Linux)
```

This is the **NetApp ONTAP MCP server**, running in Docker on the Rocky VM. It exposes ONTAP operations as MCP tools.

3.4 ONTAP Simulator / ONTAP cluster

This is the actual storage backend.

The ONTAP MCP server talks to it using the **NetApp ONTAP REST API**.

4. Data flow

When Claude is asked to fetch ONTAP information, the flow is:

```
```text id="hlx7a2"
```

Claude Desktop

-> MCP tool call

-> mcp-remote

-> ONTAP MCP server

-> ONTAP REST API on simulator/cluster

-> JSON response

-> MCP tool response

-> Claude formats the answer

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## 5. Environment used

### 5.1 Windows workstation

- Claude Desktop installed
- Node.js installed
- npm / npx available

### 5.2 Rocky Linux VM

- Rocky Linux 10.2
- Docker Engine installed
- ONTAP MCP container running on port 8083

### 5.3 ONTAP backend

- ONTAP Simulator management IP reachable from the Rocky VM
  - valid ONTAP admin credentials available for `ontap.yaml`
-

## 6. Prerequisites checklist

### 6.1 Windows prerequisites

#### Required

- Windows desktop/laptop
- internet access for downloading Claude Desktop and Node.js
- network connectivity from Windows to the Rocky VM IP and port **8083**
- Claude Desktop installed
- Node.js installed
- npm / npx available in Command Prompt

#### Why Node.js is required on Windows

Claude Desktop uses a local command to start the MCP bridge:

```
cmd id="fwv6ru" cmd /c npx -y mcp-remote http://<ROCKY_VM_IP>:8083 --allow-http --transport http-first
```

So the Windows side must have:

- Node.js
- npm
- npx

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#### # 6.2 Rocky Linux prerequisites

##### ## Required

- Rocky Linux VM
- network connectivity from Rocky VM to ONTAP simulator / ONTAP cluster management IP
- Docker Engine installed and running
- local directory for `ontap.yaml`
- firewall / routing that allows Windows to reach Rocky on port **\*\*8083\*\***

##### ## Optional but useful

- `git`
- `curl`
- `nodejs`
- `npm`

These are not mandatory for the Docker runtime itself, but they are useful for troubleshooting and testing.

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## # 7. Reference links

### ## Claude Desktop

- [Claude Desktop download page](https://claude.com/download?utm\_source=chatgpt.com)
- [Claude Desktop installation article](https://support.claude.com/en/articles/10065433-install-claude-desktop?utm\_source=chatgpt.com)

### ## Node.js

- [Node.js official download page](https://nodejs.org/en/download?utm\_source=chatgpt.com)

### ## Docker

- [Docker Engine installation docs](https://docs.docker.com/engine/install/?utm\_source=chatgpt.com)

### ## ONTAP MCP

- [NetApp ONTAP MCP documentation](https://netapp.github.io/ontap-mcp/latest/?utm\_source=chatgpt.com)

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## # 8. Installation and setup procedure

### # 8.1 Windows setup

#### # 8.1.1 Install Claude Desktop

##### ## Why

Claude Desktop is the MCP client that will invoke ONTAP MCP tools.

##### ## Steps

1. Open the Claude Desktop download page:  
[Claude Desktop download page](https://claude.com/download?utm\_source=chatgpt.com)
2. Download the Windows installer.
3. Install Claude Desktop.
4. Sign in after installation.

##### ## Verify

Launch Claude Desktop and confirm the application opens successfully.

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#### # 8.1.2 Install Node.js on Windows

##### ## Why

Node.js provides:

- `node`

```
- `npm`
- `npx`
```

`npx` is used to run `mcp-remote`, which is the local bridge Claude Desktop uses to reach the remote ONTAP MCP server.

### ## Steps

1. Open:

[Node.js official download page](https://nodejs.org/en/download?utm\_source=chatgpt.com)

2. Download the **LTS** version for Windows.

3. Run the installer.

4. Finish installation.

### ## Verify installation

Open **Command Prompt** and run:

```
``cmd id="v4jgbf"
node -v
npm -v
```

Expected: both commands return version numbers.

Example Windows result from this environment:

```
• node -v → v24.18.0
```

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## 8.1.3 Important note: use Command Prompt for testing

During setup, PowerShell blocked `npx.ps1` because script execution was disabled.

So for manual testing and validation, use **Command Prompt** rather than PowerShell.

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## 8.2 Rocky Linux setup

### 8.2.1 Verify Rocky Linux version

Run:

```
``bash id="n3s6lu" cat /etc/os-release
```

Example output:

```
```text id="r2o0jt"
NAME="Rocky Linux"
VERSION="10.2 (Red Quartz)"
```

8.2.2 Install useful packages on Rocky

Why

These tools help with troubleshooting and setup.

Install

```
```bash id="qqiviy" sudo dnf update -y sudo dnf install -y git curl tar gzip sudo dnf install -y nodejs npm
```

```
Verify
```bash id="tk5t0l"
git --version
node -v
npm -v
```

Example result from this environment:

- git version 2.52.0
- node v22.22.2
- npm 10.9.7

8.2.3 Install Docker on Rocky

Why

The ONTAP MCP server is run as a **Docker container** on the Rocky VM.

Install Docker

```
```bash id="w2b4ye" sudo dnf -y install dnf-plugins-core sudo dnf config-manager --add-repo https://download.docker.com/linux/rhel/docker-ce.repo sudo dnf makecache sudo dnf install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

```
Start and enable Docker
```bash id="v9khw1"
sudo systemctl enable --now docker
sudo systemctl status docker --no-pager
```

Verify Docker

```
```bash id="9f3vxu" docker --version sudo docker run hello-world
```

```

8.2.4 Create ONTAP MCP working directory

```bash id="mgzjlm"
mkdir -p ~/ontap-mcp
```

8.2.5 Create `ontap.yaml`

Create the configuration file:

```
```bash id="x6kl4p" vi ~/ontap-mcp/ontap.yaml
```

```
Example file
```yaml id="ifub1q"
Pollers:
  sim1:
    addr: <ONTAP_SIMULATOR_MGMT_IP>
    username: admin
    password: <ONTAP_PASSWORD>
    use_insecure_tls: true
```

Field explanation

- `sim1` → logical poller name
 - `addr` → ONTAP simulator or ONTAP cluster management IP
 - `username` / `password` → ONTAP admin credentials
 - `use_insecure_tls: true` → useful for lab or simulator environments using self-signed certificates
-

8.2.6 Start the ONTAP MCP container

Run:

```
```bash id="3v8j5o" sudo docker run -d
--name ontap-mcp-server
-p 8083:8083
-v /home/rnelluri/ontap-mcp/ontap.yaml:/opt/mcp/ontap.yaml
ghcr.io/netapp/ontap-mcp:latest
start --port 8083 --host 0.0.0.0
```

```
Why each part matters

`--name ontap-mcp-server`
Gives the container a predictable name for log and lifecycle commands.

`-p 8083:8083`
Exposes the ONTAP MCP HTTP server on the Rocky VM.

`-v /home/rnelluri/ontap-mcp/ontap.yaml:/opt/mcp/ontap.yaml`
Mounts the ONTAP configuration file into the container.

`start --port 8083 --host 0.0.0.0`
Starts the MCP server and binds it to all interfaces so Windows can reach it remotely.

9. Validation procedure

9.1 Validate ONTAP MCP on Rocky

Check the running container
```bash id="8k4wzr"
sudo docker ps
```

Expected type of output:

```
```text id="9gpnvs" CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES 15542ef1cadb
ghcr.io/netapp/ontap-mcp:latest "/ontap-mcp start -..." ... Up ... 0.0.0.0:8083->8083/tcp ontap-mcp-
server
```

```
Check that port 8083 is listening
```bash id="90v9r0"
ss -tulpn | grep 8083
```

Expected:


```
```text id="l4t9h2" tcp LISTEN 0 4096 0.0.0.0:8083 0.0.0.0:*
```

```

```

```
9.2 Validate the MCP HTTP endpoint on Rocky
```

```
Run:
```

```
```bash id="8xvqf5"
curl http://localhost:8083
```

Expected:

```
```text id="87z1gx" Bad Request: GET requires an Mcp-Session-Id header
```

```
Important interpretation
This is **expected and healthy**.
It means the MCP server is running and waiting for proper MCP requests rather
than a normal browser GET.
```

```
Additional tests used during setup:
```

```
```bash id="w7c1kc"
curl -i http://localhost:8083/
curl -i http://localhost:8083/mcp
```

These also returned the same MCP-style **400 Bad Request**, which confirmed the service was alive.

9.3 Validate ONTAP MCP logs on Rocky

Show recent logs

```
```bash id="klv3zh" sudo docker logs ontap-mcp-server --tail 100
```

```
Example successful startup log sequence:
```

```
```text id="8v98qm"
Reading config path=/opt/mcp/ontap.yaml
loaded API catalog endpoints=212
starting MCP server over HTTP transport address=0.0.0.0:8083
MCP server endpoint available url=http://0.0.0.0:8083
Server ready to accept connections
```

Follow live logs

```
```bash id="1phmht" sudo docker logs -f ontap-mcp-server
```

Use this while testing Claude Desktop to watch incoming requests and server behavior.

---

# 9.4 Identify the Rocky VM IP for Windows access

Run:

```
```bash id="td7pcg"
hostname -I
```

In this environment, the Rocky VM endpoint used by Windows was:

```
```text id="7ff45o" http://192.168.59.130:8083
```

---

# 10. Claude Desktop MCP configuration

Claude Desktop needs a local MCP server definition.  
Because the actual ONTAP MCP server is running remotely on Rocky over HTTP,  
Claude Desktop uses `mcp-remote` as a local bridge.

---

# 10.1 Standard Claude Desktop config path

Primary config file path:

```
```text id="evhqyb"
%APPDATA%\Claude\claude_desktop_config.json
```

Example resolved path:

```
```text id="hm0l22" C:\Users\ADMIN\AppData\Roaming\Claude\claude_desktop_config.json
```

---

# 10.2 Create the standard config directory if missing

If the folder does not exist, create it:

```
```cmd id="ed19gm"
mkdir "%APPDATA%\Claude"
```

Then create/edit the config file:

```
```cmd id="wrdc6h" notepad "%APPDATA%\Claude\claude_desktop_config.json"
```

```

10.3 Claude Desktop MCP config content

Paste the following:

```json id="j0zc62"
{
  "mcpServers": {
    "ontap-mcp": {
      "command": "cmd",
      "args": [
        "/c",
        "npx",
        "-y",
        "mcp-remote",
        "http://192.168.59.130:8083",
        "--allow-http",
        "--transport",
        "http-first"
      ]
    }
  }
}
```

What Claude does with this

Claude launches:

```
```cmd id="0u6w0n" cmd /c npx -y mcp-remote http://192.168.59.130:8083 --allow-http --transport http-
first
```

That process then connects to the ONTAP MCP server running on the Rocky VM.

---

### # 10.4 Important alternate Claude config path for some Windows installs

During troubleshooting, the standard `%APPDATA%\Claude` path alone was not

always sufficient for the Windows Claude Desktop install being used.

Some Windows/MSIX installations may also read or expect the config under a LocalCache package path.

```
Create this folder if needed
```cmd id="3c1nlu"
mkdir "%LOCALAPPDATA%\Packages\Claude_pzs8srxjxfjic\LocalCache\Roaming\Claude"
```

Why this may be needed

If Claude Desktop does **not** show the MCP server under **Settings → Developer → Local MCP servers** even though the JSON is correct in `%APPDATA%\Claude`, the app may be reading from its MSIX package sandbox path.

When to use this step

Use it if all of the following are true: 1. `%APPDATA%\Claude\claude_desktop_config.json` already exists 2. Claude Desktop was restarted 3. `ontap-mcp` still does not appear in the Local MCP servers list

If needed, place the same config file here too

```
```text id="ggrn6t" %LOCALAPPDATA%\Packages\Claude_pzs8srxjxfjic\LocalCache\Roaming\Claude\claude_desktop_config.json
```

```

11. Manual validation of `mcp-remote` on Windows

Before relying on Claude Desktop, validate the bridge directly.

Open Command Prompt and run:

```cmd id="n8g4mn"
npx -y mcp-remote http://192.168.59.130:8083 --allow-http --transport http-first
```

Expected successful output pattern:

```
```text id="jlwmso" Using transport strategy: http-first Using automatically selected callback port: 38030
Discovering OAuth server configuration... Connecting to remote server: http://192.168.59.130:8083
Connected to remote server using StreamableHTTPClientTransport Local STDIO server running Proxy
established successfully between local STDIO and remote StreamableHTTPClientTransport Press Ctrl+C
to exit
```

```
What this confirms
- Windows can reach the Rocky VM endpoint
- the ONTAP MCP HTTP server is reachable
- the transport works correctly
- Claude Desktop should be able to use the same bridge command
```

---

```
12. Load the MCP server in Claude Desktop
```

After saving the config:

1. Fully quit Claude Desktop
2. Reopen it
3. Go to:

```
```text id="ob7myd"
Settings -> Developer -> Local MCP servers
```

Expected result

You should see:

```
```text id="1mjlmw" ontap-mcp Running
```

That means Claude Desktop has:

- read the config file
- launched `mcp-remote`
- connected to the ONTAP MCP server successfully

---

```
13. Functional testing from Claude Desktop
```

Once the server shows as **Running**, test it with a prompt such as:

```
```text id="6m9a9h"
Using the ONTAP MCP tools, list the configured ONTAP systems and show me
basic cluster information for the simulator.
```

Other useful test prompts:

- "Using ONTAP MCP, show me cluster version, nodes, and aggregates."
- "List SVMs and volumes on the configured ONTAP simulator."
- "Show network interfaces and IPspaces from the ONTAP simulator."

14. Troubleshooting matrix

Problem / Symptom	Likely Cause	Validation	Resolution
<pre>curl http:// localhost: 8083 returns Bad Request: GET requires an Mcp- Session-Id header</pre>	Not an error; MCP server is expecting MCP protocol requests	<pre>sudo docker ps, sudo docker logs ontap-mcp-server --tail 100</pre>	Treat as healthy behavior and continue with MCP client testing
Claude Desktop does not show <code>ontap-mcp</code> under Local MCP servers	Wrong config path, JSON issue, or Claude not restarted	Check <code>%APPDATA%\Claude\claude_desktop_config.json</code> ; verify JSON; restart Claude	Fix JSON, restart Claude, and if needed place config under the LocalCache/MSIX path too
<code>%APPDATA%\Claude</code> folder does not exist	Folder not created by the app yet	Browse <code>%APPDATA%</code>	Create it manually with <code>mkdir "%APPDATA%\Claude"</code>
PowerShell blocks <code>npx.ps1</code>	PowerShell execution policy restriction	Try <code>npx</code> in PowerShell and observe the error	Use Command Prompt and configure Claude with <code>command: "cmd"</code> and <code>args: ["/c", "npx", ...]</code>
<pre>npx -y mcp-remote --version returns "Invalid URL"</pre>	<code>mcp-remote</code> expects a remote MCP URL	Reproduce from Command Prompt	Test with the actual MCP endpoint instead of <code>--version</code>
Windows can't reach the Rocky MCP endpoint	Network path, firewall, wrong Rocky IP, or server not listening on <code>0.0.0.0</code>	Run <code>npx -y mcp-remote http://<rocky-ip>:8083 --allow-http --transport http-first</code> and check Rocky logs	Confirm Rocky IP, container status, port mapping, and host binding

Problem / Symptom	Likely Cause	Validation	Resolution
Claude shows the MCP server but tool calls fail	ONTAP connectivity or credentials issue in <code>ontap.yaml</code>	Follow <code>sudo docker logs -f ontap-mcp-server</code> while issuing Claude requests	Verify ONTAP mgmt IP, credentials, and <code>use_insecure_tls</code> if using a simulator/ self-signed cert

15. Troubleshooting notes from this actual setup

15.1 Docker requirement clarification

Initially there was uncertainty about whether the Rocky Linux host itself would run the MCP server directly.

In this environment, the correct model was:

- Rocky Linux VM = Docker host
- ONTAP MCP server = Docker container

15.2 “GET bad request” was actually a good sign

Opening the endpoint from curl or a browser returned:

```
```text id="skvwp5" Bad Request: GET requires an Mcp-Session-Id header
```

That was **not** a failure. It proved the ONTAP MCP server was up and speaking MCP over HTTP.

## 15.3 PowerShell vs Command Prompt

PowerShell blocked ``npx.ps1``.

Command Prompt worked correctly, so the Claude Desktop MCP definition was intentionally built around:

```
```cmd id="pmjlmw"
cmd /c npx ...
```

15.4 Alternate Windows config path mattered

The `%LOCALAPPDATA%\Packages\Claude_pzs8srxjxfjjc\LocalCache\Roaming\Claude` path was important to document because some Windows Claude Desktop installs may use or mirror config there.

16. Operational commands

16.1 Windows commands

Verify Node.js and npm

```
```cmd id="vjlw4" node -v npm -v
```

```
Show standard Claude config
```cmd id="yd8f9w"
type "%APPDATA%\Claude\claude_desktop_config.json"
```

Create standard Claude config folder

```
```cmd id="c18exj" mkdir "%APPDATA%\Claude"
```

```
Create alternate LocalCache/MSIX Claude config folder
```cmd id="sqy8rw"
mkdir "%LOCALAPPDATA%\
Packages\Claude_pzs8srxrjxfjfc\LocalCache\Roaming\Claude"
```

Manual `mcp-remote` test

```
```cmd id="i4xrbz" npx -y mcp-remote http://192.168.59.130:8083 --allow-http --transport http-first
```

```

16.2 Rocky Linux commands

Check OS version
```bash id="lmjlmw"
cat /etc/os-release
```

Check running MCP container

```
```bash id="e2h2mk" sudo docker ps
```

```
Show recent MCP logs
```bash id="rwjy9x"
sudo docker logs ontap-mcp-server --tail 100
```


Follow live MCP logs

```
```bash id="6llhrr" sudo docker logs -f ontap-mcp-server
```

```
Check listening port
```bash id="o7j0ps"
ss -tulpn | grep 8083
```

Test local endpoint

```
```bash id="n8o5bn" curl http://localhost:8083 curl -i http://localhost:8083/ curl -i http://localhost:8083/
mcp
```

```

17. Recommended hardening / operational improvement

To make the ONTAP MCP container survive Rocky VM reboots, recreate it with a
restart policy:

```bash id="if9m13"
sudo docker rm -f ontap-mcp-server

sudo docker run -d
  --name ontap-mcp-server
  --restart unless-stopped
  -p 8083:8083
  -v /home/rnelluri/ontap-mcp/ontap.yaml:/opt/mcp/ontap.yaml
  ghcr.io/netapp/ontap-mcp:latest
  start --port 8083 --host 0.0.0.0
```

Why use `--restart unless-stopped`

If the Rocky VM reboots, Docker will automatically start the ONTAP MCP server again.

18. Final configuration summary

Rocky VM MCP endpoint

```
```text id="2o6j9j" http://192.168.59.130:8083
```

```
ONTAP MCP Docker image
```text id="lxb75r"
ghcr.io/netapp/ontap-mcp:latest
```

ONTAP MCP container name

```
```text id="fjjlwm" ontap-mcp-server
```

```
Standard Claude Desktop config path
```text id="vq6zow"
%APPDATA%\Claude\claude_desktop_config.json
```

Alternate Windows LocalCache/MSIX config path if needed

```
```text id="kzv2mr" %LOCALAPPDATA%
\Packages\Claude_pzs8srxjxfjjc\LocalCache\Roaming\Claude\claude_desktop_config.json
```

```
Final Claude Desktop config
```json id="p24c4y"
{
  "mcpServers": {
    "ontap-mcp": {
      "command": "cmd",
      "args": [
        "/c",
        "npx",
        "-y",
        "mcp-remote",
        "http://192.168.59.130:8083",
        "--allow-http",
        "--transport",
        "http-first"
      ]
    }
  }
}
```

19. Final outcome

The environment is working end-to-end:

- Claude Desktop is installed on Windows
- Node.js / npm are installed on Windows

- Claude Desktop launches `mcp-remote`
- `mcp-remote` connects to the Rocky Linux ONTAP MCP server
- the ONTAP MCP server runs in Docker on port 8083
- the MCP server reads `ontap.yaml`
- the MCP server connects to the ONTAP simulator / ONTAP cluster via ONTAP REST API
- Claude can use ONTAP MCP tools successfully and return live ONTAP data